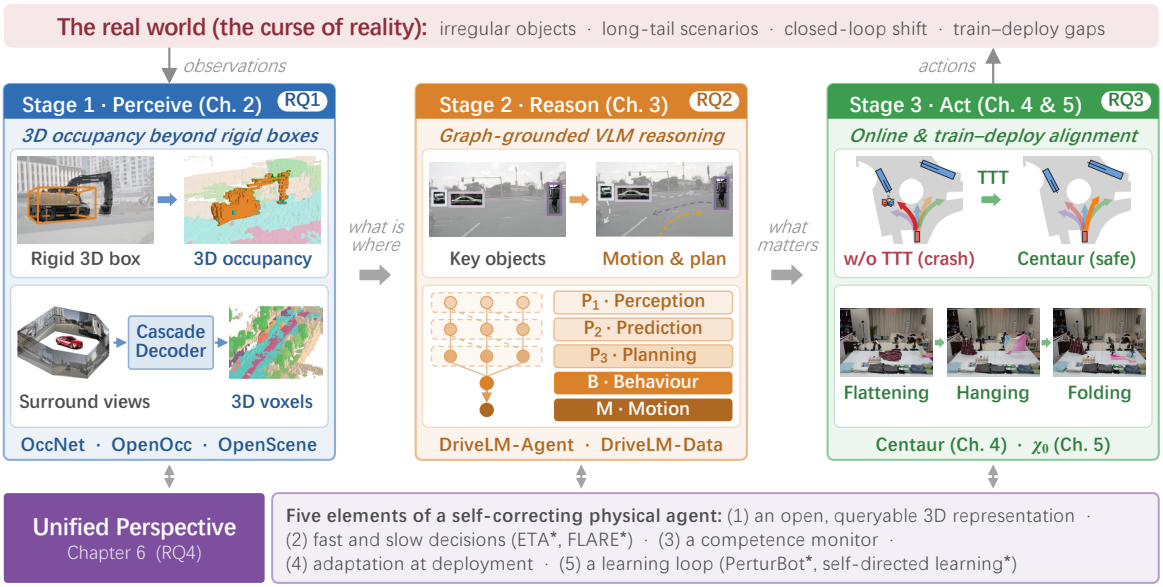

The real world (the curse of reality): irregular objects · long-tail scenarios · closed-loop shift · train–deploy gaps
observations
actions
Stage 1 · Perceive (Ch. 2)
RQ1
3D occupancy beyond rigid boxes
Rigid 3D box
3D occupancy
Cascade Decoder
Surround views
3D voxels
OccNet · OpenOcc · OpenScene
what is where
Stage 2 · Reason (Ch. 3)
RQ2
Graph-grounded VLM reasoning
Key objects
Motion & plan
P1 · Perception
P2 · Prediction
P3 · Planning
B · Behaviour
M · Motion
DriveLM-Agent · DriveLM-Data
what matters
Stage 3 · Act (Ch. 4 & 5)
RQ3
Online & train–deploy alignment
TTT
w/o TTT (crash)
Centaur (safe)
Flattening
Hanging
Folding
Centaur (Ch. 4) · χ0 (Ch. 5)
Unified Perspective
Chapter 6 (RQ4)
Five elements of a self-correcting physical agent: (1) an open, queryable 3D representation · (2) fast and slow decisions (ETA*, FLARE*) · (3) a competence monitor · (4) adaptation at deployment · (5) a learning loop (PerturBot*, self-directed learning*)